

HW0428

2026-05-01

```
library(wooldridge)
data(mroz)
# a.
# 篩選參與勞動力市場的樣本
mroz_selected <- subset(mroz, inlf == 1)
# 建立新的Dummy variables
# 若 motheduc > 12 則為 1，否則為 0
mroz_selected$MOTHERCOLL <- ifelse(mroz_selected$motheduc > 12, 1, 0)
# 若 fatheduc > 12 則為 1，否則為 0
mroz_selected$FATHERCOLL <- ifelse(mroz_selected$fatheduc > 12, 1, 0)

# 計算受過部分大學教育的父母比例
mother_college_pct <- mean(mroz_selected$MOTHERCOLL, na.rm = TRUE) * 100
father_college_pct <- mean(mroz_selected$FATHERCOLL, na.rm = TRUE) * 100
cat("ANS of a.", "\n")
```

```
## ANS of a.
```

```
cat("參與勞動樣本數:", nrow(mroz_selected), "\n")
```

```
## 參與勞動樣本數: 428
```

```
cat(sprintf("母親受過部分大學教育的百分比: %.2f%%\n", mother_college_pct))
```

```
## 母親受過部分大學教育的百分比: 12.15%
```

```
cat(sprintf("父親受過部分大學教育的百分比: %.2f%%\n", father_college_pct))
```

```
## 父親受過部分大學教育的百分比: 11.68%
```

```
# b.
vars_for_corr <- mroz_selected[, c("educ", "MOTHERCOLL", "FATHERCOLL")]
# 計算相關係數矩陣
cor_matrix <- cor(vars_for_corr, use = "complete.obs")
print(cor_matrix)
```

```
##           educ MOTHERCOLL FATHERCOLL
## educ      1.0000000  0.3594705  0.3984962
## MOTHERCOLL 0.3594705  1.0000000  0.3545709
## FATHERCOLL 0.3984962  0.3545709  1.0000000
```

```
cat("ANS of b.", "\n")
```

```
## ANS of b.
```

```
cat("EDUC 和 MOTHERCOLL 的相關係數為 0.3595 ;EDUC 和 FATHERCOLL 的相關  
係數為 0.3985 。", "\n")
```

```
## EDUC 和 MOTHERCOLL 的相關係數為 0.3595 ;EDUC 和 FATHERCOLL 的相關  
## 係數為 0.3985 。
```

```
# c.  
library(ivreg)  
wage_iv_model <- ivreg(log(wage) ~ educ + exper + I(exper^2) | MOTHERCOLL + exper + I(exper^  
2), data = mroz_selected)  
summary(wage_iv_model)
```

```
##  
## Call:  
## ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | MOTHERCOLL +  
##      exper + I(exper^2), data = mroz_selected)  
##  
## Residuals:  
##      Min      1Q   Median      3Q      Max  
## -3.08719 -0.32444  0.04147  0.36634  2.35621  
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)  
## (Intercept) -0.1327561  0.4965325  -0.267  0.78932  
## educ         0.0760180  0.0394077   1.929  0.05440 .  
## exper        0.0433444  0.0134135   3.231  0.00133 **  
## I(exper^2)   -0.0008711  0.0004017  -2.169  0.03066 *  
##  
## Diagnostic tests:  
##              df1 df2 statistic  p-value  
## Weak instruments    1 424    63.563 1.46e-14 ***  
## Wu-Hausman          1 423     0.742    0.39  
## Sargan              0 NA         NA      NA  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.6703 on 424 degrees of freedom  
## Multiple R-Squared: 0.147, Adjusted R-squared: 0.1409  
## Wald test: 8.2 on 3 and 424 DF, p-value: 2.569e-05
```

```
# 計算 95% 信賴區間  
educ_95percent_interval <- confint(wage_iv_model, level = 0.95)["educ", ]  
cat("ANS of c.", "\n")
```

```
## ANS of c.
```

```
cat("The 95% interval estimate for the coefficient of EDUC is [",  
    round(educ_95percent_interval[1], 4), ", ",  
    round(educ_95percent_interval[2], 4), "]\n")
```

```
## The 95% interval estimate for the coefficient of EDUC is [ -0.0014 , 0.1535 ]
```

```
# d.  
# 估計第一階段方程式  
first_stage_d <- lm(educ ~ exper + I(exper^2) + MOTHERCOLL, data = mroz_selected)  
# 第一階段估計結果  
summary(first_stage_d)
```

```
##  
## Call:  
## lm(formula = educ ~ exper + I(exper^2) + MOTHERCOLL, data = mroz_selected)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -7.4267 -0.4826 -0.3731  1.0000  4.9353   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept) 12.079094   0.303118  39.849  < 2e-16 ***  
## exper        0.056230   0.042101   1.336   0.182      
## I(exper^2)   -0.001956   0.001256  -1.557   0.120      
## MOTHERCOLL   2.517068   0.315713   7.973 1.46e-14 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 2.133 on 424 degrees of freedom  
## Multiple R-squared:  0.1347, Adjusted R-squared:  0.1285   
## F-statistic: 21.99 on 3 and 424 DF,  p-value: 2.965e-13
```

```
t_val <- summary(first_stage_d)$coefficients["MOTHERCOLL", "t value"]  
f_stat <- t_val^2  
  
cat("ANS of d.", "\n")
```

```
## ANS of d.
```

```
cat(sprintf("F-statistic: %.4f\n", f_stat))
```

```
## F-statistic: 63.5631
```

```
cat("MOTHERCOLL 係數的 t 值是 7.973 , F-值是 63.5631 。 F 值遠大於常用的臨界值 10 ,  
因此拒絕 MOTHERCOLL 對 EDUC 沒有影響的虛無假設 ,表示 MOTHERCOLL 是  
一個強的工具變數。另外,Stock-Yogo 臨界值是一個用於檢測工具變數強度的標  
準。如果願意接受10%的拒絕率來進行5%顯著性水平的檢定 ,則臨界值為16.38 。  
MOTHERCOLL 的 F-統計量 63.21 遠大於 16.38 ,進一步證明它是強的工具變數。", "\n")
```

```
## MOTHERCOLL 係數的 t 值是 7.973 , F-值是 63.5631 。 F 值遠大於常用的臨界值 10 ,
## 因此拒絕 MOTHERCOLL 對 EDUC 沒有影響的虛無假設 ,表示 MOTHERCOLL 是
## 一個強的工具變數。另外,Stock-Yogo 臨界值是一個用於檢測工具變數強度的標
## 準。如果願意接受10%的拒絕率來進行5%顯著性水平的檢定 ,則臨界值為16.38 。
## MOTHERCOLL 的 F-統計量 63.21 遠大於 16.38 ,進一步證明它是強的工具變數。
```

```
# e.
library(ivreg)
wage_iv_model_ef <- ivreg(log(wage) ~ educ + exper + I(exper^2) | MOTHERCOLL + FATHERCOLL + e
  xper + I(exper^2), data = mroz_selected)

summary(wage_iv_model_ef)
```

```
##
## Call:
## ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | MOTHERCOLL +
##       FATHERCOLL + exper + I(exper^2), data = mroz_selected)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.07797 -0.32128  0.03418  0.37648  2.36183
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.2790819  0.3922213  -0.712  0.47714
## educ         0.0878477  0.0307808   2.854  0.00453 **
## exper        0.0426761  0.0132950   3.210  0.00143 **
## I(exper^2)   -0.0008486  0.0003976  -2.135  0.03337 *
##
## Diagnostic tests:
##              df1 df2 statistic p-value
## Weak instruments    2 423    56.963 <2e-16 ***
## Wu-Hausman          1 423     0.519  0.472
## Sargan              1 NA      0.238  0.626
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6679 on 424 degrees of freedom
## Multiple R-Squared: 0.153, Adjusted R-squared: 0.147
## Wald test: 9.724 on 3 and 424 DF, p-value: 3.224e-06
```

```
# 計算 95% CI
educ_95percent_interval_ef <- confint(wage_iv_model_ef, level = 0.95)["educ", ]
cat("ANS of e.", "\n")
```

```
## ANS of e.
```

```
cat("The 95% interval estimate for the coefficient of EDUC using both MOTHERCOLL and FATHERCO
LL is [",
    round(educ_95percent_interval_ef[1], 4), ", ",
    round(educ_95percent_interval_ef[2], 4), "]\n")
```

```
## The 95% interval estimate for the coefficient of EDUC using both MOTHERCOLL and FATHERCOLL  
is [ 0.0273 , 0.1483 ]
```

```
library(car)
```

```
## Loading required package: carData
```

```
# f.  
# 估計第一階段方程式  
first_stage_f <- lm(educ ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL, data = mroz_selecte  
d)  
  
# 檢定 MOTHERCOLL 與 FATHERCOLL 的聯合顯著性  
f_test_result <- linearHypothesis(first_stage_f, c("MOTHERCOLL", "FATHERCOLL"))  
print(f_test_result)
```

```
##  
## Linear hypothesis test:  
## MOTHERCOLL = 0  
## FATHERCOLL = 0  
##  
## Model 1: restricted model  
## Model 2: educ ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL  
##  
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)  
## 1     425 2219.2  
## 2     423 1748.3  2    470.88 56.963 < 2.2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 建立無工具變數的限制模型  
restricted_model <- lm(educ ~ exper + I(exper^2), data = mroz_selected)  
# 建立包含工具變數的全模型  
full_model <- lm(educ ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL, data = mroz_selected)  
# F 檢定  
anova_result <- anova(restricted_model, full_model)  
print(anova_result)
```

```
## Analysis of Variance Table  
##  
## Model 1: educ ~ exper + I(exper^2)  
## Model 2: educ ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL  
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)  
## 1     425 2219.2  
## 2     423 1748.3  2    470.88 56.963 < 2.2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
cat("ANS of f.", "\n")
```

```
## ANS of f.
```

```
cat("MOTHERCOLL 和 FATHERCOLL 兩個工具變數的聯合顯著性 F-檢驗統計量為  
56.963 > 10。值也超過了Stock-Yogo臨界值,該臨界值是根據樣本大小標準來確定的。如果願意接受 10% 的拒絕  
率來進行 5% 顯著性水平的檢定,臨界值為 19.93,拒絕工具變數是弱的虛無假設。", "\n")
```

```
## MOTHERCOLL 和 FATHERCOLL 兩個工具變數的聯合顯著性 F-檢驗統計量為  
## 56.963 > 10。值也超過了Stock-Yogo臨界值,該臨界值是根據樣本大小標準來確定的。如果願意接受 10% 的  
拒絕率來進行 5% 顯著性水平的檢定,臨界值為 19.93,拒絕工具變數是弱的虛無假設。
```

```
# g.  
# 1. 資料集與模型建立  
mroz_selected$MOTHERCOLL <- ifelse(mroz_selected$motheduc > 12, 1, 0)  
mroz_selected$FATHERCOLL <- ifelse(mroz_selected$fatheduc > 12, 1, 0)  
mroz_selected$mothercoll <- mroz_selected$MOTHERCOLL  
mroz_selected$fathercoll <- mroz_selected$FATHERCOLL  
  
mroz_e_clean <- na.omit(mroz_selected)  
  
first_stage_e <- lm(educ ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL, data = mroz_e_clean)  
mroz_e_clean$educ_hat_e <- predict(first_stage_e)  
  
# 建立 iv_manual_e 模型  
iv_manual_e <- lm(log(wage) ~ educ_hat_e + exper + I(exper^2), data = mroz_e_clean)  
  
# 2. 執行 Sargan 檢定  
mroz_e_clean$residuals <- residuals(iv_manual_e)  
sargan_reg <- lm(residuals ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL, data = mroz_e_clean)  
  
r_squared <- summary(sargan_reg)$r.squared  
n <- nrow(mroz_e_clean)  
sargan_stat <- n * r_squared  
df_sargan <- 2 - 1  
p_value_sargan <- 1 - pchisq(sargan_stat, df = df_sargan)  
  
# 輸出結果  
cat("ANS of g.", "\n")
```

```
## ANS of g.
```

```
cat(sprintf("Sargan Test Statistic (LM): %.4f\n", sargan_stat))
```

```
## Sargan Test Statistic (LM): 0.2137
```

```
cat(sprintf("p-value: %.4f\n", p_value_sargan))
```

```
## p-value: 0.6439
```

```
cat("虛無假設是工具變數有效。5%的臨界值為  $0.6439 > 0.2375$ ，故無法拒絕虛無假設。", "\n")
```

```
## 虛無假設是工具變數有效。5%的臨界值為  $0.6439 > 0.2375$ ，故無法拒絕虛無假設。
```